

MACAPA, Brazil

WMO: 820980

Lat: 0.0451S

Lon: 51.1099W

Elev: 13

StdP: 101.17

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.8	22.9	21.0	15.7	31.5	21.8	16.5	29.7	8.6	29.2	7.5	29.5	1.9	0	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	9.5	35.2	26.4	35.1	26.4	34.8	26.5	28.1	32.2	27.7	32.0	27.4	31.8	5.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.0	28.7	26.9	22.6	28.7	26.2	21.7	28.6	89.7	31.7	88.1	32.7	86.8	32.5	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.1	7.2	6.4	DB	21.7	35.6	2.6	0.9	19.9	36.2	18.4	36.8	16.9	37.3	15.1	38.0
			WB	21.2	28.8	2.4	0.9	19.4	29.4	18.0	29.9	16.6	30.4	14.9	31.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	27.9	27.4	27.5	27.8	28.2	28.4	28.5	29.1	29.4	29.5	29.5	28.8
	DBStd	1.37	1.42	1.46	1.42	1.31	1.26	1.09	1.04	0.92	0.79	0.85	0.91	1.16
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6752	553	487	542	533	565	551	572	593	582	606	584	584
	CDD18.3	3710	295	254	283	283	307	301	314	334	332	347	334	326
	CDH23.3	39258	2774	2165	2435	2539	2944	3025	3246	3896	4088	4357	4168	3620
(13)	CDH26.7	16357	1016	693	752	809	1049	1179	1350	1788	1970	2142	2027	1581
(14)	Wind	WSAvg	3.2	3.2	2.9	2.8	2.5	2.5	2.6	3.3	4.0	4.2	4.1	3.7
(15) Precipitation	PrecAvg	2311	252	343	382	355	280	243	179	89	26	22	49	136
	PrecMax	2674	475	661	626	573	429	399	308	193	75	89	214	383
	PrecMin	1751	58	175	126	190	134	121	98	16	0	0	0	4
	PrecStd	232	102	112	101	98	75	69	58	45	22	26	60	100
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	34.1	34.2	34.8	35.1	35.1	35.2	35.9	36.0	36.1	36.1	35.9
		MCWB	26.4	27.0	27.2	26.9	27.1	26.8	26.5	26.4	26.4	26.3	26.0	26.2
	2%	DB	33.8	33.0	33.0	33.2	34.0	34.1	34.2	35.0	35.1	35.2	35.1	34.9
		MCWB	26.6	26.7	26.8	26.9	27.2	26.8	26.7	26.6	26.5	26.3	26.3	26.3
	5%	DB	32.8	32.0	31.9	32.0	32.9	33.1	33.2	34.1	34.2	34.8	34.8	34.0
		MCWB	26.4	26.4	26.4	26.5	26.9	26.6	26.5	26.7	26.5	26.3	26.3	26.6
	10%	DB	31.2	30.9	30.8	30.9	31.8	32.0	32.2	33.2	33.8	34.0	33.9	33.0
		MCWB	26.0	26.2	26.2	26.4	26.5	26.3	26.2	26.4	26.4	26.2	26.3	26.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	28.1	28.0	28.1	28.3	28.1	28.0	28.2	27.9	27.8	27.8	28.0
		MCDB	32.1	31.9	32.0	31.9	32.4	31.8	32.2	32.7	33.0	33.3	32.5	31.5
	2%	WB	27.1	27.2	27.2	27.4	27.7	27.5	27.4	27.6	27.3	27.2	27.2	27.3
		MCDB	31.5	31.0	30.8	30.9	31.5	31.5	31.9	32.5	32.8	33.0	32.6	31.8
	5%	WB	26.7	26.7	26.8	27.0	27.2	27.1	27.0	27.1	27.0	26.8	26.9	26.9
		MCDB	30.8	30.2	30.1	30.4	30.8	31.3	31.6	32.3	32.7	32.6	32.5	31.8
	10%	WB	26.3	26.2	26.4	26.6	26.8	26.7	26.6	26.7	26.6	26.5	26.5	26.5
		MCDB	30.0	29.3	29.2	29.5	30.1	30.5	30.8	31.6	32.0	32.1	32.0	31.2
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	7.1	7.0	7.1	7.7	8.3	8.8	9.3	9.5	9.5	9.3	8.6
		MCDBR	9.1	8.6	8.6	8.7	9.2	9.5	9.7	10.1	10.2	10.2	10.1	9.9
		MCWBR	3.0	3.1	3.1	3.0	3.1	3.2	3.4	3.2	3.0	2.9	2.9	3.0
	5% WB	MCDBR	8.2	7.9	7.8	7.8	8.3	8.7	9.2	9.5	9.7	9.7	9.5	9.1
		MCWBR	3.1	3.2	3.1	3.0	3.2	3.3	3.5	3.5	3.2	3.2	3.2	3.1
(40) Clear-Sky Solar Irradiance	taub		0.409	0.434	0.425	0.403	0.414	0.412	0.416	0.445	0.485	0.478	0.468	0.448
	taud		2.508	2.437	2.471	2.558	2.501	2.505	2.484	2.381	2.230	2.250	2.281	2.371
	Ebn at Noon		921	900	901	901	867	856	858	851	836	850	859	878
	Edh at Noon		111	121	116	104	106	104	107	122	146	144	139	126
(44) All-Sky Solar Radiation	RadAvg		4.34	4.00	4.07	4.17	4.49	4.94	5.29	5.88	6.04	5.73	5.36	4.67
	RadStd		0.30	0.25	0.24	0.20	0.27	0.32	0.18	0.21	0.20	0.26	0.27	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.82	+0.42	N/A	+1.28	+0.58	+0.81	N/A	N/A	+436
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+434

Nomenclature: See separate page